Manel MILI, PhD Student.

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About me

- I am Manel Mili, a PhD student at the Faculty of Science of Monastir and a member of the Laboratory of Medical Technology and Image Processing (LTIM) at the Faculty of Medicine, University of Monastir, Tunisia. My research focuses on artificial intelligence and computer vision applied to medical image processing. I specialize in predictive modeling, classification, and data mining, with strong expertise in deep learning, machine learning, and data visualization. Passionate about innovative problem-solving, I strive to advance methodologies that enhance decision-making and drive impactful results in research.

Employment History

2025 — **Research Internship** at ENSTA Paris - Polytechnic Institute of Paris .
- Laboratory: U2IS
- Topic: Mechanistic Interpretability applied to multimodal data for MGMT status prediction
  - Duration: 08/09/2025 →05/12/2025

2024 – 2025 — **Contract Lecturer** at Higher Institute of Informatics and Mathematics of Monastir (ISIMM) .
- Teaching module: TP – *Cloud & Virtualization*
  - Level: Software Engineering - License-3rd year
  - 01/10/2024 →31/12/2024
- Teaching module: TP – *Mobile Development*
  - Level: Software Engineering - License-3nd year
  - 01/10/2024 →31/12/2024

2023 – 2024 — **Contract Lecturer** at Faculty of Sciences of Monastir (FSM) .
- Teaching module: TP – *SOA Architecture and Web Services*
  - Level: Software Engineering and Information Systems - License-3rd year
  - 01/10/2023 →31/12/2023
- Teaching module: TP – *Database Management Systems*
  - Level: Software Engineering and Information Systems - License-2nd year
  - 01/01/2024 →31/07/2024

2022 – 2023 — **Adjunct Lecturer** at Faculty of Sciences of Monastir (FSM) .
- Teaching module: C + TD – *Data Science Introduction*
  - Level: License in Mathematics - 3rd year
  - 20/09/2022 →15/12/2022

2021 – 2022 — **Adjunct Lecturer** at Faculty of Sciences of Monastir (FSM) .
- Teaching module: C + TD – *Data Science Introduction*
  - Level: License in Mathematics - 1st year
  - 20/09/2021 →15/12/2021
- Teaching module: C + TD – *Statistical Programming with R*
  - Level: License in Mathematics - 1st year
  - 20/09/2021 →15/12/2021

Education

- 2021 – Present — **PhD Student** in FSM, Tunisia.
  - Thesis title: *Prediction of MGMT Status in Gliomas Using Artificial Intelligence*.
  - Laboratory: Medical Technologies and Imaging Laboratory - LTIM - LR12ES06
- 2017 – 2020 — **Master Degree in Software Engineering:** from ISIMM, Tunisia.
  - Memory project title: *Early Mortality Prediction in Intensive Care Units (ICUs) using Deep Learning Models*.
  - Laboratory: Medical Technologies and Imaging Laboratory - LTIM - LR12ES06
- 2012 – 2015 — **Bachelor Degree in Computer Science** from ISIMM, Tunisia.
  - End of study project title: *Distributed Optimization by Particle Testing: Case of MaxCSP*.
- 2008 – 2012 — **Baccalaureate:** Experiment Sciences at the lyceum of Jammel, Monastir, Tunisia.

Research Publications

Legend: J = Journal, C = Conference, B = Book Chapter, S = In Submission

J.1 Kriker, O., Bouchahda, N., Mili, M., Sfar, R., Ben Abdallah, A., and Bedoui, M. H. (2026). Hybrid Approach for Automatic Annotation and Displacement Estimation of Cardiac Fluid Hemodynamics. In International Journal of Imaging Systems and Technology.

J.2 Kechida, M., Slama, N., Kenani, M., Mili, M., Salem, A., Krit, W., ... and Bedoui, M. H. (2026). Machine learning models in predictive factors for megaloblastic character of macrocytic anemia. In Leukemia Research Reports, 100568.

J.3 Ben Khalifa, A., Mili, M., Maatouk, M., Ben Abdallah, A., Abdellali, M., Gaied, S., ... and Zrig, A. (2025). Deep Transfer Learning for Classification of Late Gadolinium Enhancement Cardiac MRI Images into Myocardial Infarction, Myocarditis, and Healthy Classes: Comparison with Subjective Visual Evaluation. In Diagnostics, 15(2), 207.

J.4 Salah, S., Jellad, A., Mili, M., Chaabeni, A., Bedoui, M. H., and Ben Saad, H. (2025). Comparing efficacy and adherence of smartphone-guided exercises to conventional self-directed exercises for neck pain in office workers: A randomized controlled trial protocol. In PLoS One, 20(9), e0329863.

J.5 Nouira, M., Yacoub, A., Abdallah, A. B., Mili, M., Boudriga, H., Hmida, W., ... and Chatti, K. (2025). L’IA peut-elle surseoir à la réalisation d’une scintigraphie osseuse dans le bilan initial d’un cancer de la prostate?. In Médecine Nucléaire, 49(2), 128-129.

C.1 Mili, M., Ben Abdeljelil, A., Ben Abdallah, and Bedoui, M. H. Weakly Supervised ViT-Conditioned Text Explanations in Glioma Histology. In 2026 International Joint Conference on Neural Networks (IJCNN 2026)

C.2 Mili, M., Ben Abdallah, A., and Bedoui, M. H. NdLinear-Attention Based Deep Learning Model for MGMT Promoter Methylation Prediction in Glioblastoma. In International Conference on Control, Decision and Information Technologies. (CoDIT 2026) (Accepted)

C.3 Mili, M., Ben Abdeljelil, A., Manzanera, A., Ben Abdallah, A., Otero, J. J., and Bedoui, M. H. (2026, March). Quantitatively Audited Multi-View XAI for Medical Image Classification: Application to MGMT Promoter Methylation. In 2026 International Conference on Agents and Artificial Intelligence (ICAART)

C.4 Messaoud, N. H., Zina, N., Mili, M., Dhia, R. B., Abdallah, A. B., and Bedoui, M. H. (2025, January). Feature Extraction through MS Lesion and Cerebral Lobe Segmentation for Enhanced EDSS Prediction. In 2025 International Symposium on iNnovative Informatics of Biskra (ISNIB) (pp. 1-6). IEEE.

Research Publications (continued)

C.5 Mili, M., Abdallah, A. B., Kerkeni, A., Otero, J. J., and Bedoui, M. H. (2024, November). Approche ensembliste pour la prédiction du statut MGMT chez les patients atteints de GBM. In Applications Médicales de l'Informatique: Nouvelles Approches-AMINA.

C.6 Mili, M., Abdallah, A. B., Otero, J. J., Kerkeni, A., and Bedoui, M. H. (2023, October). Revolutionizing Brain Cancer Diagnosis: Automated Prediction of MGMT Methylation Status using Histological Images. In 2023 International Conference on Cyberworlds (CW) (pp. 149-156). IEEE.

C.7 Mili, M., Kerkeni, A., Abdallah, A. B., Otero, J. J., and Bedoui, M. H. (2022, November). Gradient Boosting for Predicting MGMT Status in Gliomas. In AMINA 2022.

B Mili, M., Kerkeni, A., Ben Abdallah, A., and Bedoui, M. H. (2022, November). ICU Mortality Prediction Using Long Short-Term Memory Networks. In International Conference on Intelligent Data Engineering and Automated Learning (pp. 242-251). Cham: Springer International Publishing.

Participation in Scientific Events

- Participation to IEEE World Congress on Computational Intelligence (WCCI) - 2026: International Joint Conference on Neural Networks (IJCNN) in Maastricht, Limburg, from June 21, 2026 to June 26, 2026.
- Active member of the Tunisian Association for the Promotion of Applied Research (ATUPRA) (2024).
- Participation to STIC School - 2023: Artificial Intelligence and Medical Application, organized by *Technologies and Medical Imaging Laboratory (LTIM-LR12ES06)* in Monastir, Tunisia, from November 16, 2021 to November 18, 2021.
- Participation to IEEE CW conference event - 2023: 22th International Conference on Cyberworlds (CW2023) in Sousse, Tunisia, from October 03, 2022 to October 05, 2022.
- Participation to AMINA Workshop Event - 2022: 11th Workshop of Applications of Medical Informatics: New Approaches, organised by *Technologies and Medical Imaging Laboratory (LTIM-LR12ES06)* in Monastir, Tunisia, from November 02, 2022 to November 04, 2022.
- Participation to Biomechanics Society Congress event - 2022: 47th Congress of Biomechanics Society, organised by, *Technologies and Medical Imaging Laboratory (LTIM-LR12ES06)* in Monastir, Tunisia, from October 26, 2022 to October 28, 2022.
- Participation to Digital Festival – 2022: Monastir, Tunisia, from February 24, 2022 to February 26, 2022.
- Participation to STIC School - 2021: Artificial Intelligence and Medical Application, organized by *Technologies and Medical Imaging Laboratory (LTIM-LR12ES06)* in Monastir, Tunisia, from October 23, 2021 to November 04, 2021.
- Participation to STIC School - 2019: Implementation of a Medical Database, organized by *Technologies and Medical Imaging Laboratory (LTIM-LR12ES06)* in Monastir, Tunisia, from December 14, 2019 to December 16, 2019.

Contribution To Academic And Scientific Research

- Session Chair at IEEE World Congress on Computational Intelligence (WCCI 2026), Maastricht, Limburg (Virtual Participation), 21-26 June 2026.
  - Chair of the special session: "SS10 Trustworthy and Explainable Federated Learning: Towards Security and Privacy Future II".
  - Responsibilities included moderating presentations, managing Q&A discussions, and ensuring smooth session execution.

Contribution To Academic And Scientific Research (continued)

- Participation in the organization of the National Workshop: Applications of Medical Informatics: New Approaches (2026) , organized by *Technologies and Medical Imaging Laboratory (LTIM-LR12ES06)* in Faculty of Medicine of Monastir.
- Participation in the CEC-IA and Health program (2023), as **A TRAINER**, organized by *Technologies and Medical Imaging Laboratory (LTIM-LR12ES06)* in Faculty of Medicine of Monastir.
- Technical Reviewer in:
  - **CIBM (2024)**: Journal of Computers in Biology and Medicine (Impact Factor = 7.7).
  - **JMLR (2024)**: Journal of Machine Learning Research (`http://www.mlrjournal.org/reviewers`).
  - **AMINA (2022)**: Applications of Medical Informatics: New Approaches.
  - **ICMLA (2021)**: International Conference on Machine Learning and Applications (Impact factor = 2.97).
- Participation in the CEC-IA and Health program (2022), as **A TRAINER**, organised by *Technologies and Medical Imaging Laboratory (LTIM-LR12ES06)* in Faculty of Medicine of Monastir.
- Participation in the organization of the National Workshop:Applications of Medical Informatics: New Approaches (2022), organist by *Technologies and Medical Imaging Laboratory (LTIM-LR12ES06)* in Faculty of Medicine of Monastir.
- Participation in the organization of the International 47th Congress of the Biomechanics Society (2022), organized by *Technologies and Medical Imaging Laboratory (LTIM-LR12ES06)* in Faculty of Medicine of Monastir.

Realised Projects

- A Predictive Model for Alzheimer's Disease in Patients.
  Skills: Machine Learning · Python · Keras · Deep learning · TensorFlow
- A Project on Predicting Depression in Individuals.
  Skills: Python · Deep learning · Predictive Modeling · Data Analysis · TensorFlow
- ICU mortality prediction using deep learning.
  Skills: Predictive Analytics · Machine Learning · Big data · Artificial Intelligence (AI) · Microsoft Azure · Computer-Aided Design (CAD) · Deep learning · Predictive Modeling · Data Analysis · Data Classification · Electronic Medical Record (EMR)
- Predicting Drug Sensitivity: The development of a model for anticipating which patients who may develop drug sensitivity.
  Skills: Python · Computer-Aided Design (CAD)
- Predicting Liver Cancer: Developing an AI model capable of accurately identifying the probability of an individual developing liver cancer.
  Skills: Medical Data Interpretation · Project Planning · Deep learning · Statistical Data Analysis
- Predicting Megaloblastic Anemia.
  Skills: Artificial Intelligence (AI) · Computer Science · TensorFlow
- Predicting adverse drug reactions: Predicting adverse drug reactions, which are undesirable or harmful reactions that may occur when using medicines or drugs.
  Skills: Python · Matplotlib · Computer-Aided Design (CAD) · Keras · Pandas (Software) · TensorFlow
- Predicting the profile of exacerbators: Predicting the profile of frequent exacerbators from the first consultation onwards using clinical and para-clinical data.
- Prediction of MGMT-methylation Status for GBM Patients.
  Skills: Medical Imaging · Medical Data Interpretation · Keras · Image Processing · Deep learning · Predictive Modeling · Data Fusion · Statistical Data Analysis · Computer Science · Computer Vision · Data Classification · TensorFlow

Realised Projects (continued)

- Developed a web-based interface for predicting MGMT promoter status using histological images. The application utilizes Flask for the backend and Ngrok for secure, temporary hosting.
  - **Key Features**:
    - **Image Processing**: Integrated image upload functionality to accept histological images for analysis.
    - **Prediction Model**: Implemented a deep learning model to predict MGMT promoter status from the provided histological images.
    - **User Interface**: Designed a user-friendly interface using HTML, CSS, and JavaScript to ensure smooth interaction.
    - **Backend Framework**: Utilized Flask for server-side processing and handling HTTP requests.
    - **Secure Hosting**: Employed Ngrok to provide secure, temporary hosting and testing environment.
  - **Technologies Used**:
    - **Frontend**: HTML, CSS, JavaScript
    - **Backend**: Flask
    - **Hosting**: Ngrok
- Reducing hospital-acquired infections: Developing a model for predicting nosocomial infections for patients admitted for surgery using machine learning techniques.
  Skills: Artificial Intelligence (AI) · Medical Data Interpretation · Keras · Statistical Data Analysis · Data Classification · TensorFlow

Certifications

2026
- Certificate of participation in: ***IEEE World Congress on Computational Intelligence - International Joint Conference on Neural Networks*** *(**WCCI - IJCNN**)*.

2024
- Certificate of reviewing in: *Journal of Computer in Biology and Medicine: **CIBM***.
- Certificate of participation in: *Fighting off journal rejections on the path to manuscript acceptance - organised by: **LetPub***.

2023
- Certificate of organization of: ***Workshop of Applications of Medical Informatics: New Approaches***.
- Certificate of participation in: *Building a Career Reputation via Peeref and Other Venues - organised by: **LetPub***.
- Certificate of validation of the: **English Formation: Presentation**.
- Certificate of participation in: **Workshop** - ***Web 2.0 tools for active learning***.
- Certificate of participation in: **The 1st Pedagogical Congress**.
- Certificate of participation in: ***School STIC****: AI and Medical Applications*.
- Certificate of participation in: ***International Conference on Cyberworlds - CW***.
- Certificate of participation in: ***Publish with WILEY****: Step by Step Guide to Writing a Literature Review*.
- Certificate of participation in: ***Publish with WILEY****: Scientific Writing Tips for non-native English Speakers*.

2022
- Certificate of participation in: *International Conference on Intelligent Data Engineering and Automated Learning - **IDEAL***.
- Certificate of participation in: *Workshop of Applications of Medical Informatics: New Approaches - **AMINA***.

Certifications (continued)

- Certificate of organization of: ***Congress of the Biomechanics Society***.
- Certificate of participation in: ***Congress of the Biomechanics Society***.
- Certificate of participation in: **Digital Festival** - *Digital Technology and its Applications*.

2021
- Certificate of participation in: ***School STIC:*** *AI and Medical Applications*.
- Certificate of attendance in: ***Article writing and bibliographic management (60 min)***. Certificate of reviewing in: *International Conference on Machine Learning and Applications :* ***ICMLA***.

Key Skills

Languages
- Strong reading, writing and speaking competencies for English (**Toeic certification 2019)**, French, Arabic and little for Espagnol.

Coding
- Java, Python, R, SQL, XML/XSL, LaTeX, Ruby, MATLAB.

Databases
- MySQL, PostgreSQL, HBase, Elasticsearch, Oracle Database, MongoDB, Microsoft SQL Server.

Web Dev
- HTML, CSS, JavaScript, Apache Web Server, Tomcat Web Server.

Artificial Intelligence
- Deep Learning, Machine Learning:
  Tools: Tensorflow, Keras, PyTorch, Scikit-learn, Theano, Caffe, XGBoost, LightGBM, Fastai, The Microsoft Azure Machine Learning SDK.

Misc.
- Academic research, teaching, training, consultation, LaTeX typesetting and publishing.

Character Working

- Strong leadership, communications and collaboration
- Creative thinking
- Analytical and critical thinker
- Problem solver and innovator
- Decision-making
- Time management
- Continuous learning
- Effective delegation

Hobbies

- Playing sports (basketball, tennis, etc.)
- Playing chess and solving puzzle games
- Reading and writing books and articles
- Traveling
- Computing

Personal Interests

- Organizing events in the community
- Being involved with charities
- Volunteering at local companies, clubs and organizations

Personal Interests (continued)

- Attending conference events